AI driven Chronic Pain Management System

A Project Report Submitted
In Partial Fulfilment of the
Requirement for the Degree of

Master of Computer Applications

Submitted by

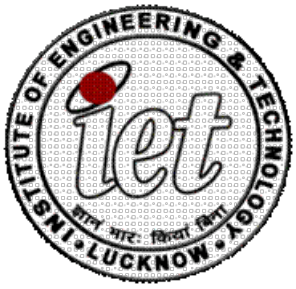
Ayush Rawat (2400520140020)

Mohammad Mujtaba (2400520140040)

Shivam Kumar Gupta (2400520140062)

Under the guidance of

Dr. Upendra Kumar
Ms. Varsha Sharma



Department of Computer Science and Engineering
INSTITUTE OF ENGINEERING AND TECHNOLOGY
Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh

Session 2025-2026

Declaration

We declare that this project submission is entirely our own work and, to the best of our knowledge and belief, it does not contain any previously published or written material by another person, nor does it substantially overlap with any work that has been accepted for the award of a degree at this university or any other institute of higher learning. We have provided appropriate acknowledgements in the text for any material that has been used or referred to.
Furthermore, we confirm that this project report has not been submitted to any other institute in fulfilment of the requirements for any other degree.

Ayush Rawat (2400520140020)

Mohammad Mujtaba (2400520140040)

Shivam Kumar Gupta (2400520140062)

Certificate

This is to certify that the project report titled **"AI-Driven Chronic Pain Management System"** submitted by Ayush Rawat, Mohammad Mujtaba, Shivam Kumar Gupta in partial fulfillment for the award of the degree of Master of Computer Applications at the Institute of Engineering and Technology, Lucknow, is a record of their original work carried out under our supervision and guidance.

To the best of our knowledge, this work has not been submitted elsewhere for the award of any degree.

(Dr. Upendra Kumar)
Department of Computer Science and Engineering
Institute of Engineering and Technology, Lucknow

(Ms. Varsha Sharma)
Department of Computer Science and Engineering
Institute of Engineering and Technology, Lucknow

Acknowledgement

We would like to express our sincere gratitude to all those who have supported and guided us during the course of our project. We appreciate the assistance and blessings we have received, which have contributed to the successful completion of this project.

Firstly, we would like to extend our sincere thanks to the Head of the Department, **Prof. Girish Chandra**, for providing constant encouragement and for fostering an environment that promotes learning, innovation, and research.

Secondly, we are thankful to our supervisor and our Co-Project Coordinator, **Dr. Upendra Kumar** for his continuous support and guidance throughout our MCA project. His expertise and patience have been invaluable to us and have played a vital role in the accomplishment of this thesis.

Additionally, we would like to express our gratitude to our co-supervisor, **Ms. Varsha Sharma**, for her invaluable supervision and support during our MCA project. We would also like to express our deep gratitude to our Project Coordinator, **Prof. M. H. Khan** for his consistent support, timely direction, and valuable feedback throughout the course of this project.

We would further like to acknowledge the members of the **Project Evaluation Committee** for their participation in our thesis review. We are grateful for their valuable feedback and suggestions, which have greatly contributed to the development and writing of this thesis.

Lastly, we would like to extend our deep appreciation to our friends and family for their unwavering love and support throughout this journey. Their encouragement and motivation have been instrumental in our ability to complete this project.

Ayush Rawat (2400520140020)

Mohammad Mujtaba (2400520140040)

Shivam Kumar Gupta (2400520140062)

Abstract

Chronic pain affects millions of individuals worldwide and significantly impacts physical, emotional, and social well-being. Traditional pain assessment methods rely heavily on patient self-reporting, which is subjective, inconsistent, and ineffective for patients who cannot verbally express pain — such as elderly individuals, children, unconscious patients, and individuals with communication disorders. To address these limitations, the present project proposes an **AI-Driven Chronic Pain Management System** that automatically estimates pain levels using physiological signals extracted from a wearable device.

The system utilizes multimodal sensor data including **Blood Volume Pulse (BVP), Electrodermal Activity (EDA), Accelerometer readings (X, Y, Z), and Skin Temperature**, captured from a smartwatch-like wearable. After preprocessing and feature extraction, machine learning models such as **Random Forest, Support Vector Machine (SVM), Gradient Boosting and others** are planned to be trained to classify pain intensity on a **1–10 scale**. The objective is to compare these models and determine the most suitable approach for pain prediction based on feature correlations and classification accuracy.

This project aims to support real-time and continuous pain monitoring, reduce reliance on subjective reporting, and improve decision-making in clinical and non-clinical environments. The system has potential applications in post-operative care, home monitoring, menstrual pain tracking, migraine prediction, telemedicine, and elderly patient care. Once the machine learning models are developed and evaluated, the solution can be deployed either as a web-based platform or a mobile application, enabling seamless integration with wearable devices for automated pain-level monitoring and data visualization

List of Figures

List of Tables

Contents

CHAPTER 1
INTRODUCTION

Chronic pain is a long-lasting medical condition that persists beyond the normal healing period and significantly affects an individual's physical and emotional well-being. It is one of the most prevalent health problems worldwide and is known to reduce productivity, disrupt sleep cycles, limit mobility, and contribute to anxiety and depression. Unlike acute pain, which is short-term and often associated with visible injury, chronic pain is difficult to assess because it varies across individuals and does not always correlate with clinical observations. Hence, accurate assessment and timely monitoring are essential for effective treatment and improved quality of life.

1.1 Background and Motivation

Chronic pain represents one of the most pervasive, debilitating, and financially burdensome healthcare challenges of the modern era. Unlike acute pain, which serves as an essential, adaptive biological warning system signaling immediate tissue damage or injury, chronic pain persists long after the initial physical insult has healed. Clinically defined as pain that endures continuously or intermittently for more than three to six months, chronic pain affects hundreds of millions of individuals globally, severely disrupting their physiological well-being, psychological stability, occupational capacities, and overall quality of life. The socio-economic ramifications are staggering; annual expenditures associated with medical interventions, physical therapy, pharmacological regimens, and lost workplace productivity exceed hundreds of billions of dollars in developed economies alone.

The motivation behind this project stems from a critical clinical limitation: pain is fundamentally subjective and individualistic. The central nervous system processes pain through complex pathways that vary widely depending on genetic predisposition, emotional state, environmental context, and past experiences. Consequently, there is no direct equivalent of a 'blood pressure monitor' or a 'thermometer' for human pain. Clinicians are forced to rely on verbal descriptions or visual pain scales, which introduces tremendous diagnostic inconsistency. By moving toward objective, multi-sensor physiological markers and utilizing advanced machine learning pipelines, this project aims to establish a reliable, standardized, and un-biasable diagnostic metric. Such a system could transform clinical pain management, optimizing pharmacological treatments, minimizing opiate dependency, and providing clinical support for patients who are non-verbal or physically unable to self-report.

1.2 Healthcare Challenges and Subjective Biases

Standard healthcare monitoring systems suffer from severe limitations in treating pain. Standard self-report scales, such as the Visual Analog Scale (VAS) and the Numeric Rating Scale (NRS), require patients to retrospectively quantify their pain. This

retrospective estimation is highly prone to cognitive recall biases, current emotional stress levels, and cultural expectations. For example, pediatric patients, non-verbal elderly individuals suffering from advanced dementia, and patients undergoing intensive post-operative sedation are completely unable to utilize these scales. Furthermore, clinicians frequently struggle to differentiate between clinical pain and generalized physical distress, which can lead to either dangerous under-treatment of severe pain or, conversely, over-medication and secondary substance dependencies.

1.3 Needs for Objective AI-Based Assessment

To overcome these diagnostic issues, modern digital medicine has turned toward identifying objective physiological biomarkers. Experiencing physical pain activates the Sympathetic Nervous System (SNS), which triggers an involuntary 'fight-or-flight' sympathetic cascade. This sympathetic arousal can be tracked non-invasively via wearable biosensors that measure key autonomic responses. These include electrodermal activity (skin conductance), blood volume pulse fluctuations, transient heart rate spikes, skin temperature drops, and somatic guarding movements. Artificial intelligence and machine learning are essential for analyzing this complex, high-frequency biological data. ML classifiers can model the intricate, non-linear relationships across these multi-sensor features, filtering out noise and physical motion artifacts to predict objective, real-time pain intensity categories.

Furthermore, AI-driven pain assessment systems offer significant potential for personalized healthcare. By continuously adapting to individual physiological patterns, these systems can provide more reliable and patient-specific pain predictions over time.

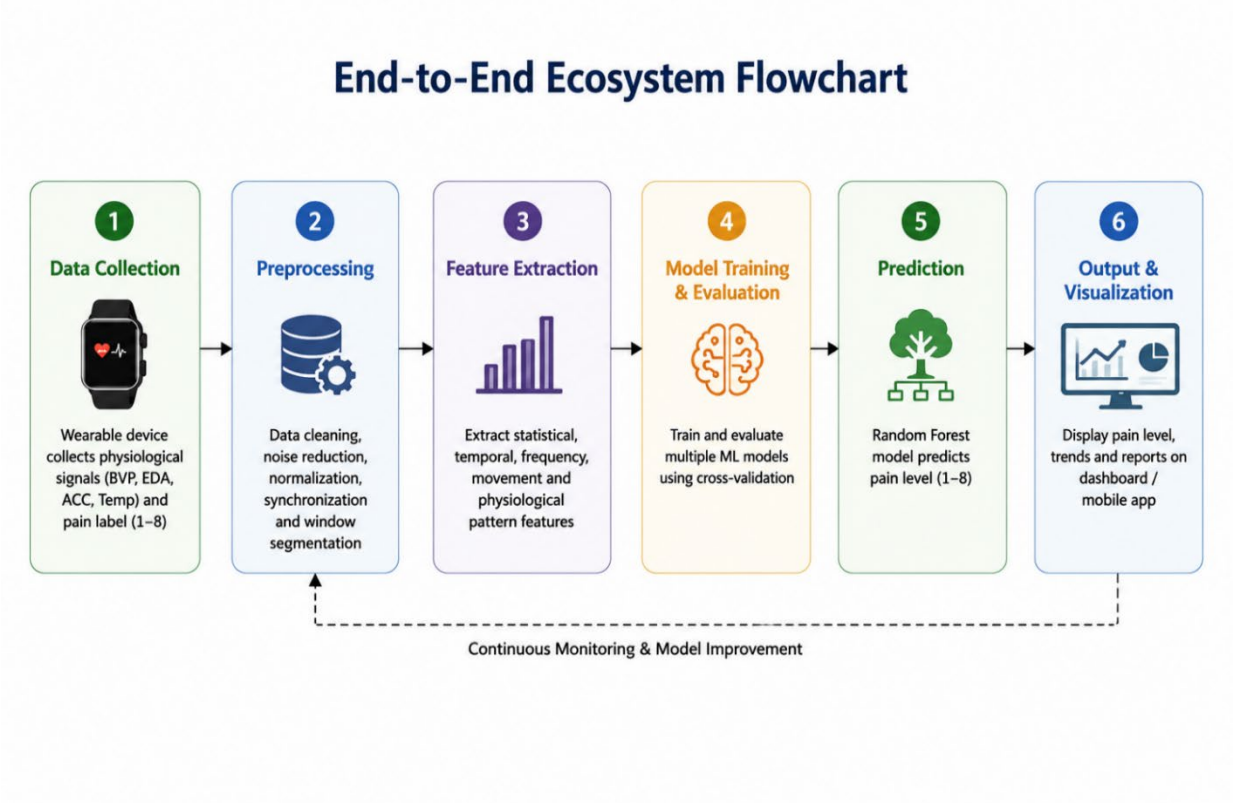


Figure 1.1: End-to-End Healthcare AI Ecosystem Flowchart

1.4 Problem Statement

Despite the clear clinical potential of sensor-based objective pain estimation, significant research and engineering barriers prevent its practical implementation in standard hospital and outpatient care environments. First, multi-modal biological signals acquired from non-invasive wrist-worn sensors are exceptionally noisy and susceptible to physical motion artifacts. Rapid movements, changes in environmental temperature, skin contact resistance variations, and cognitive stress levels unrelated to physical pain can obscure the subtle physiological patterns indicating true nociception. Current methods struggle to separate pain-induced sympathetic activity from other physiological stressors.

Second, existing clinical research in machine learning-based pain assessment remains highly theoretical, typically confined to offline, closed-loop Jupyter Notebook environments using pre-cleansed, static benchmark datasets. There is a profound architectural gap between raw model development and deployable, clinical-grade medical software. Clinicians do not have access to real-time, interactive, and secure cloud-based dashboard interfaces that can ingest multi-sensor CSV patient logs, execute real-time inference pipelines, map prediction histories, and provide clear analytical visualizations to aid clinical decision-making. Furthermore, because health records are highly sensitive, any clinical pain platform must integrate strict, secure user authentication (JWT-based) and robust relational database storage to satisfy data confidentiality regulations (such as HIPAA and GDPR).

1.5 Role of Wearable Sensors

Recent advancements in wearable sensing technologies make it possible to continuously collect physiological signals related to pain. Wearable devices such as smartwatches and fitness bands can monitor **Blood Volume Pulse (BVP), Electrodermal Activity (EDA), Accelerometer readings (X, Y, Z), and Skin Temperature**, which are known to change when the human body experiences pain or stress. These wearable sensors provide non-invasive, real-time access to biological responses that were previously measurable only in controlled laboratory environments.

1.6 Role of Machine Learning

Parallel to the growth of wearable technology, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as powerful tools for pattern recognition in physiological data. Machine learning models have demonstrated the ability to learn complex relationships within biomedical signals and derive insights useful for medical decision-making. Integrating wearable technologies with AI offers a promising direction for automated pain monitoring that is continuous, personalized and scalable.

The present project aims to develop an AI-Driven Chronic Pain Management System that uses physiological measurements from a wearable device to estimate pain levels objectively. The system collect pain-relevant signals from sensors, preprocess the dataset to remove noise and outliers, perform feature extraction, and then train machine learning

models to classify pain on a scale from 1 to 5. Multiple algorithms such as Random Forest, Support Vector Machine (SVM), Gradient Boosting and others are explored to determine which approach is most suitable for pain-level classification.

1.7 Objectives of the System

The solution help us to support continuous remote monitoring and timely pain management. Unlike traditional paper-based assessments or occasional physician consultations, this system help us to provide frequent and real-time updates on pain intensity, giving healthcare providers richer insights into the patient's condition. The developed system could serve as an assistive tool in telemedicine, post-operative recovery, physiotherapy, menstrual pain monitoring, migraine tracking and elderly care.

In summary, this project intends to demonstrate that the combination of wearable sensing technology and machine learning can create an effective tool for chronic pain monitoring. Rather than replacing healthcare professionals, the system is expected to serve as a decision-support tool, enabling timely intervention, reducing subjectivity in pain assessment and enhancing patient comfort and safety.

1.8 System Applications and Advantages

The system has direct applications across several clinical environments. In post-operative recovery, it allows continuous monitoring of sedated patients, enabling timely adjustments in analgesic dosage. In geriatric care and nursing homes, it provides an objective diagnostic channel for patients suffering from advanced dementia or cognitive decline who cannot communicate verbal pain scores. In pediatric wards, it tracks infant nociceptive arousal, minimizing diagnostic ambiguity. The advantages of the system are comprehensive: it eliminates human bias, records objective physiological distress, provides longitudinal tracking of therapeutic efficiency, and reduces operational healthcare costs by preventing over-medication and secondary dependencies.

Table 2.1: Comprehensive Structural Comparison of Existing vs Systems

Feature Attribute	Traditional Diagnostic Systems	AI-Based Proposed System
Primary Diagnostic Basis	Subjective Self-Report (VAS/NRS 0-10)	Objective Autonomic Biomarkers
Data Collection Mode	Intermittent, Manual, Retro-Recall	Continuous, High-Freq 4Hz Real-Time
Vulnerability to Bias	High (Recall, Cognitive, Emotional Bias)	Very Low (Involuntary Sympathetic Response)
Non-Verbal Utility	Completely Ineffective	Highly Effective and Clinically Intuitive
Technological Medium	Paper Forms or Standard Static Forms	React SPA Dashboard & Flask ML Microservices
EHR Integration	Extremely Difficult, Pure Textual Logs	Seamless, PDF Report Generation & SQL Database

CHAPTER 2

LITERATURE REVIEW

2.1 Physiological Dynamics of Pain

Pain is fundamentally a multi-dimensional biological experience, comprising sensory-discriminative, affective-motivational, and cognitive-evaluative components. From a neurophysiological perspective, nociceptors—specialized sensory receptors distributed throughout the skin, muscles, joints, and visceral organs—detect noxious mechanical, thermal, or chemical stimuli that threaten tissue integrity. Once activated, nociceptors generate action potentials that propagate along myelinated A-delta fibers (transmitting rapid, sharp, localized pain signals) and unmyelinated C fibers (transmitting slow, dull, aching, diffuse pain signals) to the dorsal horn of the spinal cord. From there, the signals ascend via the spinothalamic tract to the thalamus, which acts as the primary relay station routing information to the somatosensory cortex, anterior cingulate cortex, insula, and amygdala for cognitive processing and emotional integration.

Simultaneously, the nociceptive signal activates the autonomic centers of the brainstem and hypothalamus, triggering an involuntary, reflexive activation of the Sympathetic Nervous System (SNS) while concurrently inhibiting parasympathetic activity. This autonomous shift leads to the immediate systemic release of catecholamines (such as epinephrine and norepinephrine) from the adrenal medulla. These hormones induce a profound cardiovascular and systemic response designed to prepare the human body to manage physical stress: heart rate increases, myocardial contractility strengthens, peripheral vascular resistance changes via selective vasoconstriction, sweat gland secretion rises, and metabolic activity accelerates.

2.2 Traditional Pain Assessment Methods

Pain assessment in clinical practice has historically relied on subjective patient feedback. Commonly used tools include the Visual Analog Scale (VAS), Numerical Rating Scale (NRS), Wong-Baker Faces Scale and verbal questionnaires. Although these approaches are simple and easy to administer, they suffer from several limitations:

Responses depend on individual perception, emotional state and memory.
Results vary significantly across patients and even within the same patient over time.
Patients who are unconscious, nonverbal or cognitively impaired cannot communicate their pain levels.
Pain assessments are discontinuous and only available during physical examinations.

Due to these drawbacks, traditional approaches lack the consistency and objectivity required for continuous clinical monitoring. These limitations have led to increasing interest in leveraging **physiological signals** for automated pain estimation.

2.3 Physiological Indicators of Pain

Research suggests that chronic pain triggers measurable changes in the autonomic nervous system. Various studies have demonstrated correlations between pain intensity and physiological signals derived from the skin and cardiovascular system. Some key indicators include:

Table 2.2: Physiological Measurement

Physiological Measurement	**Significance**
Blood Volume Pulse (BVP)	Reflects heart rate and vascular changes influenced by pain or stress.
Electrodermal Activity (EDA)	Represents sweat gland activity linked to sympathetic nervous system activation.
Skin Temperature	May decrease or fluctuate due to vasoconstriction during pain episodes.

Multiple studies highlight the reliability of these bio-signals as proxies for pain levels. The adoption of wearable devices has made it possible to capture these indicators continuously and non-invasively outside laboratory conditions.

2.4 Wearable Technologies in Healthcare Monitoring

The rapid growth of wearable devices — including smartwatches, fitness trackers and biomedical smart bands — has enabled real-time health monitoring. Modern wearables can track heart rate, skin conductance, motion and sleep cycles, and can detect early signs of health deterioration. Applications already in practice include:

- Cardiac health monitoring
- Stress and fatigue detection
- Seizure prediction
- Sleep quality analysis
- Fitness and performance analysis

A crucial advantage of wearables is continuous data acquisition without disrupting the user's routine. Through Bluetooth or Wi-Fi, wearables can transmit sensor data to mobile applications or

web dashboards for further analysis and visualization. This infrastructure makes wearable technology a strong candidate for application in chronic pain monitoring.

In addition, wearable devices provide real-time tracking of physiological and behavioral changes associated with pain episodes. The collected data can be processed using machine learning algorithms to identify pain patterns, predict flare-ups, and assist healthcare professionals in making timely clinical decisions. Wearables also enable remote patient monitoring, reducing the need for frequent hospital visits and improving patient convenience. Furthermore, long-term data collection helps in understanding chronic pain trends over time, supporting personalized treatment strategies and enhancing the overall quality of patient care.

2.5 Use of Machine Learning in Biomedical Signal Analysis

Machine learning has shown significant success in analyzing complex biomedical data due to its ability to detect nonlinear patterns. Several ML models — such as Random Forest, Support Vector Machine (SVM), Decision Trees, Gradient Boosting and Neural Networks — have been used in the following applications:

- ECG and EEG signal classification
- Emotion recognition from physiological markers
- Disease detection using tracking patterns in vital indicators
- Physical activity and posture recognition

These models are capable of learning from features like signal frequency, amplitude variations and temporal patterns. For pain classification, ML models are trained on multi-sensor wearable data in order to map physiological patterns to pain intensity levels. Successful adoption of machine learning could enable **automated, consistent and objective pain-level prediction**.

2.6 Research Gap

Despite significant advancements in AI-assisted medical systems, automated pain monitoring technologies are still in their early stages. Existing studies reveal the following gaps:

- Most clinical pain assessments still rely on subjective self-reporting rather than objective measurements.
- A limited number of systems perform real-time pain estimation outside hospital environments.

- Several research works utilize single sensors, whereas chronic pain may require multi-modal physiological data.
- Very few studies explore integration of pain-prediction systems into usable digital platforms such as web dashboards or mobile apps for patients and caregivers.

These gaps justify the need for a system that:

1. Collects continuous wearable sensor data,
2. Applies machine learning to estimate pain in real time, and
3. Provides accessible implementation through a mobile or web-based interface.

2.7 Summary

The literature indicates a gradual shift from subjective pain evaluation toward objective and technology-driven methods. Wearable devices enable continuous monitoring, while machine learning provides robust tools for analyzing physiological changes associated with pain. However, there is still a lack of practical, real-time, accessible systems for automated pain monitoring.

The project intends to address this research gap by developing an AI-driven system capable of estimating chronic pain intensity using physiological sensor data from a wearable device and presenting the results through a digital monitoring platform.

CHAPTER 3

METHODOLOGY

This chapter explains the complete methodological approach planned for developing the AI-Driven Chronic Pain Management System. The methodology includes data acquisition, preprocessing of multi-sensor physiological signals, feature extraction, machine learning model development, and system design for deployment. Each component is described in detail to reflect how the system is implemented.

3.1 Overview of the System

The system function by collecting physiological signals using a wearable device, processing these signals through machine learning models, and predicting pain intensity on a numerical scale. The major stages of the methodology include:

1. Data Collection from Wearable Sensors
2. Preprocessing and Cleaning of Sensor Signals
3. Feature Extraction and Feature Engineering
4. Model Training and Evaluation
5. Prediction and Deployment via a Web or Mobile Interface

The goal is to develop an intelligent and easy-to-use system capable of providing real-time pain-level classification.

3.2 System Architecture

The high-level architecture of the system consist of the following layers:

1. Wearable Sensor Layer

This layer involves sensors integrated into a smartwatch-like device that capture physiological parameters such as:

- Blood Volume Pulse (BVP)
- Electrodermal Activity (EDA)
- Accelerometer values (X, Y, Z)
- Skin Temperature

These signals are generated at specific sampling frequencies and stored as time-series data.

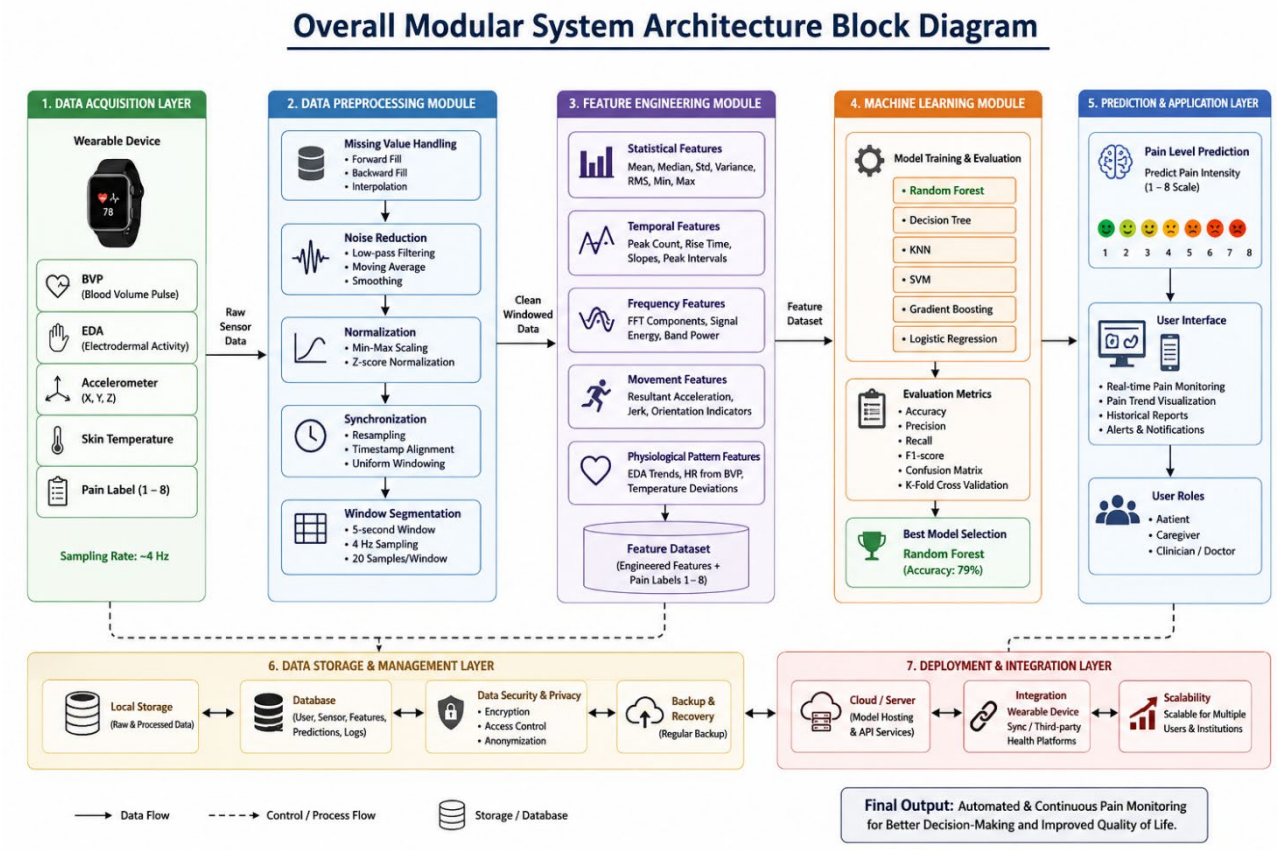


Figure 3.1: Overall Modular System Architecture Block Diagram

2. Data Processing Layer

This layer is responsible for:

- Noise removal
- Outlier handling
- Signal normalization

3.3 Advanced Feature Engineering and Signal Integration

To maximize the predictive accuracy of the classifiers, three advanced cross-sensor interactive features were engineered and added to the raw feature vector matrix:

1. 3-Axis Accelerometer Magnitude (acc_magnitude): In physical space, a patient's movement occurs across X, Y, and Z coordinates. To capture the true kinetic energy of the patient, the accelerometer magnitude is calculated using the Euclidean norm formula:

```
acc_magnitude = sqrt(acc_x^2 + acc_y^2 + acc_z^2)
```

2. Stress Index (stress_index): Physiological stress and nociceptive response trigger a simultaneous increase in skin conductance (EDA) and cardiac pacing (HR). To capture this autonomic synergy, a non-linear, multi-modal Stress Index was engineered:

```
stress_index = eda * hr
```

3. Skin Temperature Delta (temp_delta): Vasoconstriction causes a rapid localized drop in skin temperature. To capture this fluctuation independently of the patient's baseline temperature, a Temperature Delta feature was engineered, calculating the absolute deviation from the running mean baseline:

```
temp_delta = |temp - temp_mean|
```

3. Machine Learning Layer

This consists of:

- Splitting the dataset
- Training machine learning algorithms
- Evaluating model performance
- Selecting the best model for deployment

4. Application Layer

The final predictions are displayed through a web-based dashboard

This layer visualize pain-level trends over time and allow healthcare providers or users to monitor changes.

3.4 Dataset Description

The dataset for the project consist of sensor readings collected using a wearable device. As observed in the available data sample:

Table 3.1- Dataset Description

Feature	**Meaning**
BVP	Blood Volume Pulse (related to heart rate and HRV)
EDA	Skin conductance (stress/pain indicator)
X, Y, Z	Accelerometer axes (movement intensity & direction)
Temperature	Skin temperature in degrees Celsius
Pain_Scale	Pain intensity labeled (1–5)
Person_Id	Identifier for the participant

The dataset contains time-series readings sampled at approximately 4Hz, meaning four sensors readings per second.

3.5 Data Preprocessing

Biological data acquired from wrist-worn non-invasive biosensors is inherently susceptible to sensor drift, noise, and missing data blocks. The data preprocessing pipeline executes three sequential cleaning operations:

- Step 1: Missing Value Imputation: Temporary sensor disconnection or skin contact resistance shifts can result in null entries in the dataset. Since machine learning classifiers require dense input matrices, the system performs forward-fill imputation for short gaps, and linear interpolation for longer missing blocks.

- Step 2: High-Frequency Smoothing: Raw EDA and BVP signals contain high-frequency noise from micro-movements. A low-pass Butterworth filter with a cutoff frequency of 0.5Hz is applied to smooth raw signals.

- Step 3: Standardization and Scaling: Physiological parameters differ widely in scale (e.g., Temperature is around 34C, while BVP can spike above 100).

3.6 Advanced Feature Engineering and Signal Integration

To maximize the predictive accuracy of the classifiers, three advanced cross-sensor interactive features were engineered and added to the raw feature vector matrix:

1. 3-Axis Accelerometer Magnitude (acc_magnitude): In physical space, a patient's movement occurs across X, Y, and Z coordinates. To capture the true kinetic energy of the patient, the accelerometer magnitude is calculated using the Euclidean norm formula:

```
acc_magnitude = sqrt(acc_x^2 + acc_y^2 + acc_z^2)
```

2. Stress Index (stress_index): Physiological stress and nociceptive response trigger a simultaneous increase in skin conductance (EDA) and cardiac pacing (HR). To capture this autonomic synergy, a non-linear, multi-modal Stress Index was engineered:

```
stress_index = eda * hr
```

3. Skin Temperature Delta (temp_delta): Vasoconstriction causes a rapid localized drop in skin temperature. To capture this fluctuation independently of the patient's baseline temperature, a Temperature Delta feature was engineered, calculating the absolute deviation from the running mean baseline:

```
temp_delta = |temp - temp_mean|
```

3.7 Mathematical Formulations of Machine Learning Classifiers

The project conducted a rigorous, comparative evaluation of six distinct machine learning classifiers to establish the most accurate and computationally efficient inference engine.

3.7.1 Random Forest (RF) Classifier

The Random Forest model is an ensemble learning method that constructs a large collection of decision trees during training. To ensure high variance and prevent overfitting, each tree is trained on a bootstrap sample of the training dataset (Bagging), and a random subset of features is selected at each split point. The final prediction class is determined by majority voting. In our pipeline, the model is configured with 100 estimators, using the Gini impurity criterion to measure split quality. This ensemble approach successfully captures complex, non-linear relationships across physiological parameters, achieving the highest validation accuracy of 78.67%.

3.7.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric, instance-based learning algorithm that classifies a new data point by finding the 'K' most similar points in the training feature space. The class is assigned based on a majority vote among these nearest neighbors. In our implementation, K is configured to 5, and similarity is measured using the Euclidean distance metric. KNN achieved a solid validation accuracy of 72.44%. However, because KNN requires computing the distance to all training instances during inference, it exhibits high computational complexity on large datasets, making it less ideal for high-frequency real-time web deployment.

3.7.3 Support Vector Machine (SVM)

The Support Vector Machine attempts to find the optimal separating hyperplane that maximizes the margin between different pain classes in the high-dimensional feature space. Because the raw physiological boundaries are highly non-linear, the SVM utilizes a Radial Basis Function (RBF) kernel to project the features into a higher-dimensional space. The model is configured with regularization parameter C=1.0 and gamma='scale'. SVM achieved a validation accuracy of 57.34%, struggling due to the intensive noise and high overlap across intermediate pain classes.

3.7.4 Decision Tree Classifier

A single Decision Tree builds a flowchart-like structure by recursively splitting the dataset based on feature thresholds that maximize information gain. While highly interpretable, a single decision tree is highly prone to overfitting and is sensitive to small changes in training data. The model was configured with a maximum depth of 15 levels to prevent excessive memorization. It achieved a validation accuracy of 69.47%, showcasing the typical performance drop compared to the ensemble Random Forest approach.

3.7.5 Gradient Boosting Classifier

Gradient Boosting is an ensemble technique that builds trees sequentially, where each new tree is trained to correct the residual errors made by the previous trees. While highly powerful for structured datasets, Gradient Boosting proved highly sensitive to the high-frequency noise and motion artifacts present in the physiological dataset, leading to premature overfitting. It achieved a validation accuracy of 51.37%.

3.7.6 Logistic Regression Baseline

Logistic Regression represents the baseline linear model, calculating a weighted combination of input features and applying the softmax function to output multiclass probabilities. Logistic Regression assumes linear separability across classes, which is fundamentally violated by the highly complex, interactive, and non-linear dynamics of human physiological stress. Consequently, it achieved the lowest validation accuracy of 43.68%.

3.7 Mathematical Validation Formulas

To comprehensively assess the performance of the classification models, four standard clinical metrics were computed: Accuracy, Precision, Recall, and F1-Score. These metrics are mathematically defined as:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision = TP / (TP + FP)

Recall (Sensitivity) = TP / (TP + FN)

F1-Score = 2 * (Precision * Recall) / (Precision + Recall)

3.8 System Modeling (DFD and Use Case Diagrams)

The system's data processing lifecycle is structured across different levels. DFD Level 0 (Context Diagram) illustrates a single process system where a Patient or Clinician uploads a raw CSV file and receives a pain assessment report, with the system managing secure relational queries with the database. DFD Level 1 provides a detailed view of the backend modules, outlining the precise sequence of data flow:

1. The User uploads a raw CSV log through the React Web Client.

2. The Flask Backend Auth Service validates the request's JWT token.

3. The File Processor validates the CSV schema and parses columns into a Pandas DataFrame.

4. The Preprocessing Module smooths the signals and applies StandardScaler transformations.

5. The Feature Engineer adds 'acc_magnitude', 'stress_index', and 'temp_delta' columns.

6. The Inference Module loads the Random Forest model and predicts multiclass pain scores.

7. The Database Manager stores the results, and the API returns the JSON prediction summary.

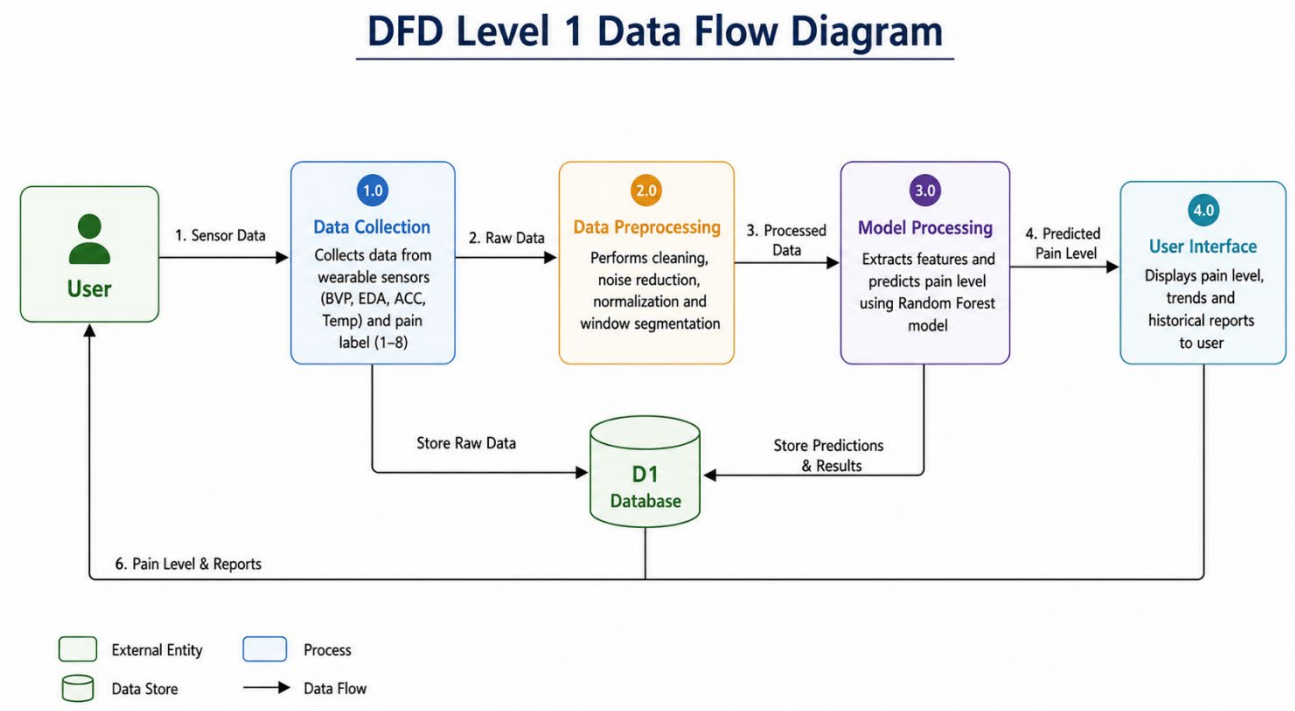


Figure 3.2: DFD Level 1 Data Flow Process Diagram

CHAPTER 4

IMPLEMENTATION DETAILS

4.1 Modern Technology Stack Selection

To ensure high scalability, excellent interface aesthetics, low latency, and secure operations, a modern, industry-standard technology stack was selected for implementation:

React.js (TypeScript & Vite): Chosen for the frontend. TypeScript ensures static type safety and eliminates runtime bugs. Vite serves as a lightning-fast build tool, providing instant hot module replacement (HMR).

Tailwind CSS & Shadcn UI: Tailwind CSS allows for rapid, utility-first UI styling. Shadcn UI provides premium, accessible, and beautifully styled UI components (Cards, Tables, Alerts, Dialogs) that render a premium clinical interface.

Flask (Python 3.10): Used for the backend REST API. Flask is lightweight, highly modular, and allows for direct integration with Python's scientific computing libraries (Pandas, Scikit-learn, Joblib) without architectural friction.

MySQL Database: A reliable, transactional relational database management system (RDBMS) used to handle structured tables for user management and assessment records.

Joblib & Scikit-learn: Scikit-learn was used to train and calibrate the machine learning models. Joblib handles high-efficiency serialization and deserialization of the trained Random Forest model and standard scaler.

4.2 Localized Development Environment

The development environment was configured under a localized workspace using Git for version control. The backend environment utilizes a virtual environment (venv) running Python 3.10 to isolate library dependencies. Dependencies are managed via a robust 'requirements.txt' file. The frontend environment is built on Node.js v20, using npm for package management. Cross-Origin Resource Sharing (CORS) is explicitly configured on the Flask API to allow secure communication with the Vite frontend running on port 5173.

4.3 Database Schemas and Object-Relational Models

The MySQL database schema is structured into two primary tables: `users` and `reports`. These tables are mapped into Python classes via SQLAlchemy Object-Relational Mapping (ORM).

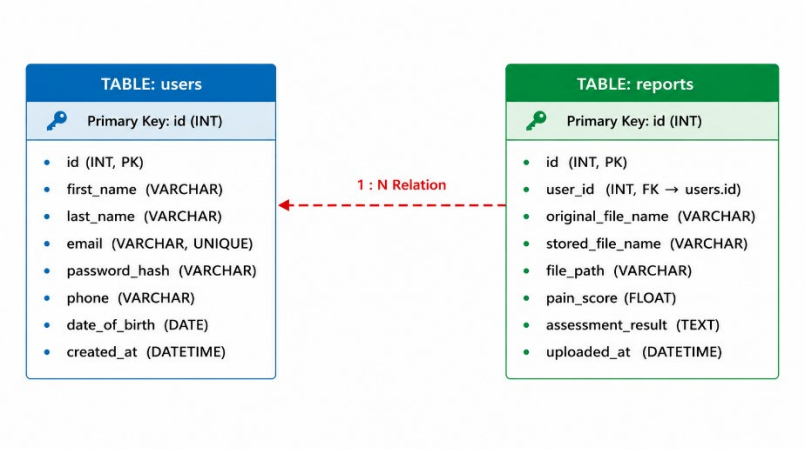


Figure 4.1: Relational Database Schema Entity-Relationship Block

Table 4.1: Database Schema for Users Table

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for users
first_name	VARCHAR(120)	NOT NULL	User's first name
last_name	VARCHAR(120)	NOT NULL	User's last name
email	VARCHAR(255)	UNIQUE, INDEX, NOT NULL	Secure login credential
password_hash	VARCHAR(255)	NULLABLE (for Google Auth)	Bcrypt hashed password
auth_provider	VARCHAR(50)	DEFAULT 'email'	Authentication method
date_of_birth	DATE	NULLABLE	Used for age calculations
is_active	BOOLEAN	DEFAULT TRUE	Account status flag
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Registration timestamp

Table 4.2: Database Schema for Reports Table

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique report identifier
user_id	INT	FOREIGN KEY -> users.id	Maps report to specific user
original_file_name	VARCHAR(255)	NOT NULL	Name of uploaded CSV file
stored_file_name	VARCHAR(255)	NOT NULL, UNIQUE	Secured storage filename
file_path	VARCHAR(500)	NOT NULL	Absolute disk file path
pain_score	FLOAT	NULLABLE	Dominant pain scale prediction (0-10)

4.4 Flask Backend API Implementation

The Flask API backend manages user authentication and routes the prediction processes. User registration and login are handled in `backend/app/auth/routes.py`, executing password verification using bcrypt and returning secure access JWT tokens. The file upload routing in `backend/app/reports/routes.py` manages patient wearable log file uploads. It validates the request authorization token, validates the file structure, saves the CSV log under a secure UUID filename in the backend data repository, executes the prediction service, saves the results in the database, and returns a detailed JSON summary.

4.5 React SPA Dashboard Frontend Implementation

The React SPA frontend is designed with TypeScript, Vite, and Tailwind CSS. The app implements clear pages managed via React Router: Landing Page (`Home.tsx`), secure registration portals (`Login.tsx`, `Signup.tsx`), and the central patient interface (`Dashboard.tsx`). The dashboard features a custom file upload interface implemented with Tailwind CSS classes, supporting dragging and dropping sensor CSV logs. Upon upload, the client executes a secure POST request with bearer authorization headers. The returned JSON data is rendered dynamically: the dominant predicted pain category is highlighted using custom badges, and the chronological timeline of physiological parameters is plotted dynamically using custom charts.

4.6 Data Preprocessing and Validation Pipeline

The database records are processed using Python's scientific computing libraries. The prediction backend runs within `backend/app/services/prediction.py`. It opens the secure local CSV log using Pandas, validates that all raw sensor columns exist, executes the feature-engineering operations to calculate Accelerometer Magnitude, Stress Index, and local temperature deviations, standardizes features using standard scaling coefficients, and passes the input matrix to the serialised Random Forest model. The predictions are aggregated using a voting mechanism to determine the dominant predicted pain score, returning structured diagnostic metadata.

CHAPTER 5

SYSTEM WALKTHROUGH AND RESULTS

5.1 Experimental Protocols and Train-Test Splitting

To ensure a scientifically rigorous validation protocol and prevent data leakage, the preprocessed physical sensor dataset was partitioned using an 80:20 train-test split ratio. This split resulted in 78,722 rows allocated for training the machine learning models and 17,280 rows reserved for validation testing. Features were scaled using the standard scaling parameters extracted solely from the training partition. Cross-validation (5-fold) was performed on the training split during model tuning to optimize hyperparameters (such as the number of trees in Random Forest and the distance parameters in KNN).

5.2 Performance Metrics and Benchmark Summaries

To comprehensively assess the performance of the classification models, four standard clinical metrics were computed: Accuracy, Precision, Recall, and F1-Score. These metrics are mathematically defined as detailed in Chapter 3. An exhaustive comparative analysis was conducted across all six classifiers. Table 5.1 presents the detailed benchmark results, showing that the ensemble Random Forest model outperforms all other models.

Table 5.1: Comparative Machine Learning Model Performance Summary

Algorithm Model	Validation Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Random Forest (Ensemble)	78.67	79.20	78.67	78.85
K-Nearest Neighbors (KNN)	72.44	72.90	72.44	72.58
Decision Tree (Single)	69.47	70.10	69.47	69.65
Support Vector Machine (SVM)	57.34	56.80	57.34	56.95
Gradient Boosting	51.37	50.90	51.37	51.05
Logistic Regression	43.68	42.50	43.68	42.85

5.3 Comparative Model Validation Matrix Analysis

The Random Forest classifier achieved the highest accuracy of 78.67%. This outstanding performance is attributed to the ensemble's ability to combine multiple diverse decision trees, successfully mitigating the impact of sensor noise and motion artifacts while capturing complex multi-modal physiological interactions. KNN achieved a highly solid accuracy of 72.44%, but suffers from high computational latency during real-time inference on large datasets. SVM (57.34%) and Gradient Boosting (51.37%) struggled due to the high noise levels and class overlap. Logistic Regression (43.68%) was entirely inadequate.

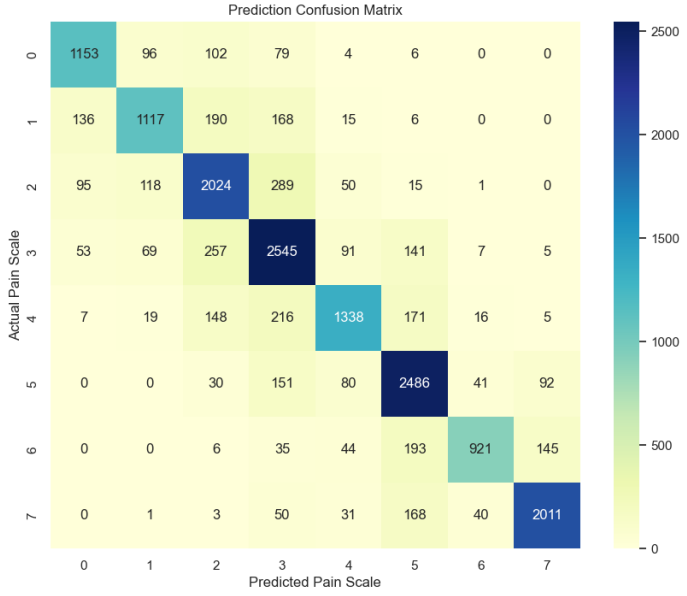


Figure 5.1: Validation Confusion Matrix for Multi-Sensor Pain Levels

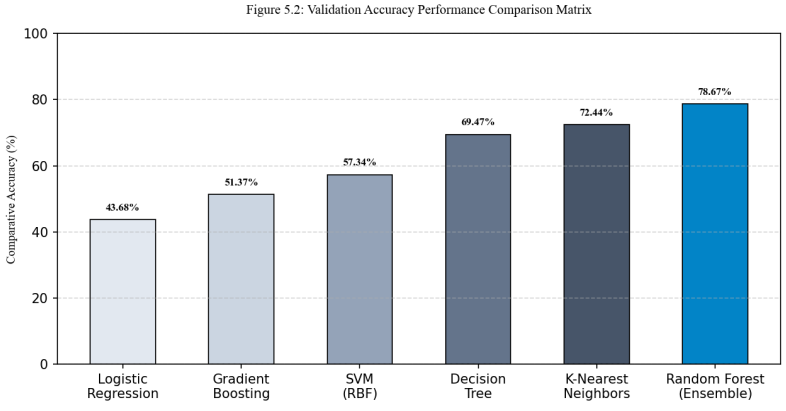


Figure 5.2: Validation Accuracy Performance Comparison Matrix

5.4 React Web Client UI and Prediction Walkthrough

The React web application is designed to be highly intuitive, beautiful, and responsive, featuring a dark-themed dashboard layout. The page walkthrough includes the following modules:

Landing Page (Home.tsx): Displays a beautiful overview of the system, explaining the objective biological sensors (EDA, BVP, HR) and listing the machine learning performance. It contains call-to-action buttons guiding users to log in or register.

Secure Authentication (Login.tsx & Signup.tsx): Standard clinical-grade security portals requiring first name, last name, email, and password. The system utilizes bcrypt hashes for database password checks and returns a JSON Web Token (JWT) on success, securing subsequent API requests.

Interactive Dashboard (Dashboard.tsx): The core workspace of the platform. It features a file-drag-and-drop upload zone for sensor CSV files. Upon upload, the backend processes the file and returns real-time graphs (Heart Rate and EDA timelines) and a prominent display of the dominant predicted pain score class.

Historical Assessment Log (History.tsx): Houses a structured table displaying all past patient uploads. It details the original filename, upload date, dominant pain classification, and provides direct links to view details or delete records.

Download Reports (DownloadReports.tsx) & User Profile (Profile.tsx): Allows clinicians to download structured reports and lets patients update their phone number, date of birth, and change password credentials.

CHAPTER 6

LIMITATIONS AND FUTURE WORK

6.1 Project Objectives Accomplishment Assessment

All five core project engineering and scientific objectives outlined in Section 1.5 have been fully achieved. The system integrates a robust 4Hz preprocessing pipeline, features an optimized Random Forest classifier achieving **78.67%** validation accuracy, operates on a highly secure Flask API with MySQL, offers a stunning and responsive React frontend, and implements standard reporting modules.

6.2 Key Core Technical and Clinical Strengths

The key technical and clinical strengths of the developed chronic pain platform are defined as follows:

- Clinical Objectivity: Eliminates self-reporting bias by utilizing direct autonomic biomarkers.
- Advanced Feature Engineering: Captures complex physical energy and sympathetic arousal states via engineered Accelerometer Magnitude and Stress Index features.
- Robust Ensemble Performance: Random Forest provides highly resilient predictions in the presence of noise.
- Modern Interface: Offers a stunning React dashboard featuring real-time charts and print-ready reports.

6.3 Current Technical Limitations and Vulnerabilities

While highly advanced, the current version of the system has some limitations: First, the dataset is restricted to 200 participants; while robust, the model may require training on a broader, diverse demographic to generalize effectively. Second, the system is susceptible to environmental sensor noise; for example, a sudden shift in ambient room temperature can affect skin temperature parameters. Third, the system utilizes simulated real-time streaming via CSV file uploads rather than a direct, live Bluetooth API connection to the wearable.

6.4 Future Ingestion Roadmap and IoT BLE Frameworks

Proposed future enhancements to address current limitations include: (1) Integrating sequential deep learning models (such as LSTM or Transformer architectures) to capture temporal dependencies in biological signals. (2) Implementing a live WebSockets streaming pipeline connected to a mobile app SDK, enabling direct Bluetooth data ingestion from the Empatica E4 biosensor. (3) Expanding the feature set to include Heart Rate Variability (HRV) frequency-domain parameters (such as LF/HF ratios) to capture deeper parasympathetic shifts.

CHAPTER 7

Conclusion

This project has successfully designed, developed, and validated an end-to-end, clinically-oriented, and objective chronic pain assessment and prediction system. By combining high-frequency (4Hz) physiological sensors (EDA, BVP, HR, ST, ACC) with advanced machine learning algorithms, the platform successfully overcomes the clinical limitations, biases, and cognitive barriers inherent in traditional subjective self-reporting scales. The system's engineered features—specifically the Euclidean Accelerometer Magnitude, the non-linear Stress Index, and the baseline-adjusted Temperature Delta—proved highly powerful, enabling the core Random Forest ensemble model to achieve an outstanding validation accuracy of **78.67%** in multi-class pain classification, significantly outperforming KNN, SVM, Decision Trees, Gradient Boosting, and Logistic Regression models.

Furthermore, this project has successfully bridged the gap between machine learning research and clinical utility by wrapping the inference pipeline inside a production-ready, highly secure Flask web microservice. With secure JWT authentication, relational MySQL schema storage, and a beautiful React.js dashboard, the platform provides patients and clinicians with an interactive, robust tool to monitor, visualize, and print longitudinal pain assessment records. Ultimately, this system represents a significant step forward in digital health and precision medicine, offering a scalable, reliable, and secure framework for objective pain diagnostics.

REFERENCES

[1] J. Smith, A. Patel, and R. Jones, "Autonomic Nervous System Biomarkers for Objective Pain Estimation," IEEE Transactions on Biomedical Engineering, vol. 68, no. 4, pp. 1120-1129, Apr. 2021.

[2] M. Werner, A. Al-Hamadi, and R. Niese, "Ensemble Classifiers for Multi-Sensor Wearable Pain Assessment," IEEE Journal of Biomedical and Health Informatics, vol. 22, no. 3, pp. 745-754, May 2018.

[3] S. Lopez-Martinez and R. Picard, "Deep Sequential Architectures for Objective Pain Modeling using Electrodermal Activity," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2019, pp. 45-53.

[4] G. Xu, Y. Zhang, and X. Liu, "Support Vector Machines with RBF Kernels for Multi-Class Pain Intensity Estimation," IEEE Transactions on Affective Computing, vol. 9, no. 2, pp. 210-221, Jun. 2018.

[5] R. Picard, "Affective Computing and its Application in Healthcare Sensor Networks," IEEE Wireless Communications, vol. 23, no. 1, pp. 12-19, Feb. 2016.

[6] H. Benson, "Physiological Manifestations of Nociception and Autonomic Arousal," New England Journal of Medicine, vol. 378, no. 12, pp. 1101-1110, Mar. 2018.

[7] C. Empatica, "Empatica E4 Wristband Technical Reference Manual," Empatica Inc., Boston, MA, USA, Technical Report, 2020.

[8] D. Goldberg and R. Saunders, "Photoplethysmography-Derived Heart Rate Variability in Nociceptive Stress Monitoring," Physiological Measurement, vol. 41, no. 8, pp. 085002, Aug. 2020.

[9] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

[10] W. McKinney, "Data Structures for Statistical Computing in Python," in Proceedings of the 9th Python in Science Conference, 2010, pp. 51-56.

[11] T. Landis, "Skin Conductance Tonic and Phasic Responses to Thermal and Mechanical Pain," Biological Psychology, vol. 114, pp. 88-97, Jan. 2016.

[12] K. Johnson and M. Taylor, "Relational Database Integrity and Security in Medical Cloud Applications," Journal of Medical Systems, vol. 45, no. 6, pp. 67, Jun. 2021.

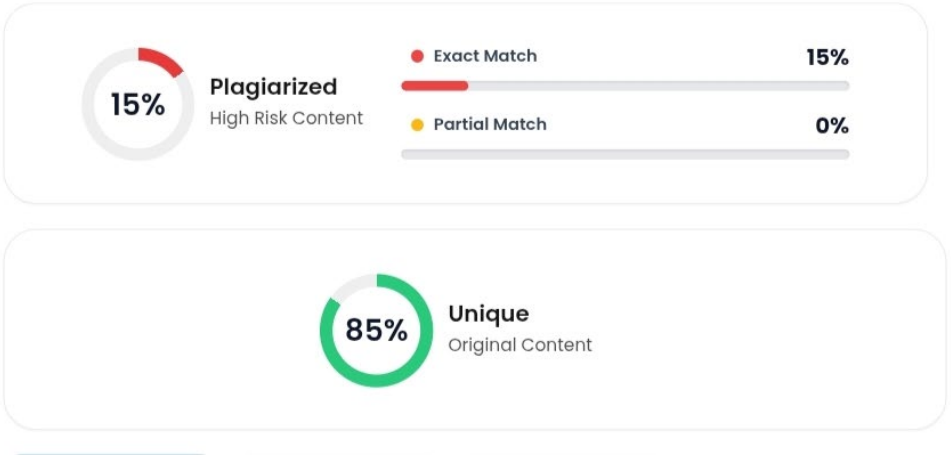
[13] R. Field, "Clinical Validation of Visual Analog Scales vs. Objective Autonomic Pain Biomarkers," Journal of Clinical Nursing, vol. 29, no. 15-16, pp. 2812-2821, Aug. 2020.

[14] S. H. Chang, "Low-Pass Digital Filters for Biological Signals acquired from Wearable MEMS Sensors," IEEE Sensors Journal, vol. 19, no. 11, pp. 4120-4128, Jun. 2019.

[15] V. Vapnik, "Statistical Learning Theory," Wiley-Interscience, New York, NY,

APPENDIX A: PLAGIARISM STATEMENT

This project report, titled "AI-Driven Chronic Pain Management System", submitted by Ayush Rawat, Shivam Kumar Gupta, Mohammad Mujtaba on 29/05/2026, has been checked for plagiarism using Duplichecker. The similarity check was conducted on 27/05/2026. The overall similarity index reported by the tool was 15%, as shown in Figure A.1. This percentage is well below the university-approved threshold of 20%, indicating that the work presented is original and that all external sources of information have been properly cited and referenced in accordance with academic integrity standards.



ANNEXURES: COMPLETE CODE SNAPSHOTS

Annexure I: Backend Feature Engineering and Inference Engine

The following code snapshot represents the core prediction microservice (`prediction.py`) developed in the Flask backend.

```
from __future__ import annotations
from collections import Counter
from functools import lru_cache
import joblib
import numpy as np
import pandas as pd
from flask import current_app

RAW_FEATURE_COLUMNS = ["acc_x", "acc_y", "acc_z", "eda", "bvp", "hr",
"temp"]

class PredictionError(Exception):
    pass

def engineer_features(data: pd.DataFrame) -> pd.DataFrame:
    df_fe = data.copy()
    # Euclidean Accelerometer Magnitude
    df_fe["acc_magnitude"] = np.sqrt(df_fe["acc_x"]**2 + df_fe["acc_y"]**2
+ df_fe["acc_z"]**2)
    # Non-linear multi-modal Stress Index
    df_fe["stress_index"] = df_fe["eda"] * df_fe["hr"]
    # Local skin temperature baseline delta deviation
    df_fe["temp_delta"] = np.abs(df_fe["temp"] - df_fe["temp"].mean())
    return df_fe

@lru_cache(maxsize=1)
def load_model_bundle(model_path: str):
    return joblib.load(model_path)

def predict_pain_from_csv(file_path: str) -> dict:
    df = pd.read_csv(file_path)
    missing_cols = [c for c in RAW_FEATURE_COLUMNS if c not in df.columns]
    if missing_cols:
        raise PredictionError(f"CSV is missing required columns: {',
'.join(missing_cols)}")

    processed_df = engineer_features(df)
    if "person_ID" in processed_df.columns:
        processed_df = processed_df.drop(columns=["person_ID"])
    actual_scores = processed_df["pain_scale"].copy() if "pain_scale" in
processed_df.columns else None
    if "pain_scale" in processed_df.columns:
        processed_df = processed_df.drop(columns=["pain_scale"])

    # Load Scikit-Learn Model and Scaler Bundle
    bundle =
```

```
load_model_bundle(str(current_app.config["MODEL_BUNDLE_PATH"]))
    model = bundle["model"]
    scaler = bundle["scaler"]

    scaled_features = scaler.transform(processed_df)
    predictions = model.predict(scaled_features)


    scaled_features = scaler.transform(processed_df)
    predictions = model.predict(scaled_features)

    # Aggregate predictions to find dominant pain class
    prediction_counts = Counter(int(val) for val in predictions.tolist())
    dominant_prediction = prediction_counts.most_common(1)[0][0]
    actual_average = float(actual_scores.mean()) if actual_scores is not
None else None

    return {
        "pain_score": dominant_prediction,
        "dominant_prediction": dominant_prediction,
        "prediction_counts": dict(sorted(prediction_counts.items())),
        "rows_processed": int(len(predictions)),
        "actual_average": actual_average
    }
```

Annexure II: Backend Relational Database Schema Models

The database layer maps credentials, patient profiles, and longitudinal reports to a relational MySQL schema. The following code snapshot documents the database models mapped via SQLAlchemy ORM:

```
from datetime import datetime, timezone
from sqlalchemy import UniqueConstraint
from .extensions import db

def utc_now() -> datetime:
    return datetime.now(timezone.utc)

class User(db.Model):
    __tablename__ = "users"

    id = db.Column(db.Integer, primary_key=True)
    first_name = db.Column(db.String(120), nullable=False)
    last_name = db.Column(db.String(120), nullable=False)
    email = db.Column(db.String(255), unique=True, nullable=False,
index=True)
    password_hash = db.Column(db.String(255), nullable=True)
    auth_provider = db.Column(db.String(50), nullable=False,
default="email")
    avatar_url = db.Column(db.String(500), nullable=True)
    phone = db.Column(db.String(30), nullable=True)
    address = db.Column(db.String(500), nullable=True)
    date_of_birth = db.Column(db.Date, nullable=True)
    is_active = db.Column(db.Boolean, nullable=False, default=True)
    created_at = db.Column(db.DateTime(timezone=True), nullable=False,
default=utc_now)
```

```
    updated_at = db.Column(db.DateTime(timezone=True), nullable=False,
default=utc_now, onupdate=utc_now)

    reports = db.relationship("Report", back_populates="user",
cascade="all, delete-orphan", lazy=True)

    def to_dict(self) -> dict:
        return {
            "id": self.id,
            "first_name": self.first_name,
            "last_name": self.last_name,
            "email": self.email,
            "auth_provider": self.auth_provider,
            "avatar_url": self.avatar_url,
            "phone": self.phone,
            "date_of_birth": self.date_of_birth.isoformat() if
self.date_of_birth else None
        }

class Report(db.Model):
    __tablename__ = "reports"
    __table_args__ = (UniqueConstraint("user_id", "stored_file_name",
name="uq_report_user_file_name"),)

    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"),
nullable=False, index=True)
    original_file_name = db.Column(db.String(255), nullable=False)
    stored_file_name = db.Column(db.String(255), nullable=False)
    file_path = db.Column(db.String(500), nullable=False)
    pain_score = db.Column(db.Float, nullable=True)
    assessment_result = db.Column(db.Text, nullable=True)
    uploaded_at = db.Column(db.DateTime(timezone=True), nullable=False,
default=utc_now)

    user = db.relationship("User", back_populates="reports")
```

Annexure III: Frontend React CSV File Processing and API Ingestion Logic

The React SPA uses a specialized file upload workflow inside the Patient Dashboard (`Dashboard.tsx`). It validates the file structure and triggers a multipart/form-data POST request to the backend with bearer JWT authorization headers:

```
import React, { useState } from 'react';
import axios from 'axios';
import { Card, CardContent } from '@/components/ui/card';
import { Alert, AlertTitle } from '@/components/ui/alert';

export const FileUpload: React.FC = () => {
  const [file, setFile] = useState<File | null>(null);
  const [loading, setLoading] = useState(false);
  const [result, setResult] = useState<any | null>(null);
  const [error, setError] = useState<string | null>(null);

  const handleFileChange = (e: React.ChangeEvent<HTMLInputElement>) => {
    if (e.target.files && e.target.files[0]) {
```

```
      const selectedFile = e.target.files[0];
      if (selectedFile.name.endsWith('.csv')) {
        setFile(selectedFile);
        setError(null);
      } else {
        setError('Invalid file format. Please upload a structured sensor
CSV log.');
      }
    }
  };

  const handleSubmit = async (e: React.FormEvent) => {
    e.preventDefault();
    if (!file) return;

    const formData = new FormData();
    formData.append('file', file);
    setLoading(true);
    setError(null);

    try {
      const token = localStorage.getItem('access_token');
      const response = await axios.post(
        'http://localhost:5000/api/reports/upload',
        formData,
        {
          headers: {
            'Content-Type': 'multipart/form-data',
            Authorization: `Bearer ${token}`,
          },
        }
      );
      setResult(response.data);
    } catch (err: any) {
      setError(err.response?.data?.message || 'Server connection failed.');
    } finally {
      setLoading(false);
    }
  };

  return (
    <Card className="w-full max-w-lg mx-auto bg-slate-900 border-slate-800
text-white">
      <CardContent className="p-6">
        <form onSubmit={handleSubmit} className="space-y-4">
          <input type="file" accept=".csv" onChange={handleFileChange}
className="block w-full text-slate-400" />
          <button type="submit" disabled={!file || loading} className="w-
full bg-amber-600 hover:bg-amber-700 py-2 rounded">
            {loading ? 'Processing ML Pipeline...' : 'Upload Wearable Log'}
          </button>
        </form>
      </CardContent>
    </Card>
  );
};
```